

ETHICS OF OFFSHORING: NOVO NORDISK AND CLINICAL TRIALS IN EMERGING ECONOMIES

Professor Klaus Meyer wrote this case solely to provide material for class discussion. The author does not intend to illustrate either effective or ineffective handling of a managerial situation. The author may have disguised certain names and other identifying information to protect confidentiality.

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On a warm day in early spring 2008, the telephone is ringing in the office of Anders Dejgaard, chief medical officer of Novo Nordisk, a leading developer and manufacturer of insulin and related products. A business journalist of the Danish national newspaper *Berlingske Tidende* is on the line and asking for an interview. Dejgaard knows her from several conversations relating to business practices in the pharmaceutical industry.

The journalist is investigating the offshoring of clinical trials by Danish companies. A report recently published in the Netherlands alleges that multinational pharmaceutical companies routinely conduct trials in developing countries under allegedly unethical conditions. Also, the Danish National Committee on Biomedical Research Ethics has expressed concerns because Danish pharmaceutical companies are not obtaining ethical reviews in Denmark for such trials despite the offer from this committee. Thus, she wants to discuss Novo Nordisk's position on these issues.

Dejgaard reflects on how to react. Several articles on ethical aspects related to medical research in the Third World had appeared in the Danish press in recent months, creating an atmosphere of suspicion towards the industry.¹ Should he meet with the journalist and if so, what should he tell her? Or should he rather focus on his forthcoming business trip to new production facilities and send Novo Nordisk's press officer to meet the journalist? In his mind flashes the possibility of derogatory headlines in the tabloid press. As a company emphasizing corporate responsibility, the interaction with the media presents both opportunities and risks to Novo Nordisk.

¹ See in particular B. Alfter, "De fattige er verdens nye forsøgskaniner. Krav om kontrol med medicinalindustrien," *Information*, Feb. 26, 2008, pp. 4-5, and B. Lambeck and S.G. Jensen, "Halvdelen af al medicin afprøves i den tredje verden," *Politiken*, Oct. 6, 2007.

NOVO NORDISK²

Novo Nordisk A/S had been created in 1989 through a merger between two Danish companies, Novo Industri A/S and Nordisk Gentofte A/S. Both had been established in the 1920s as manufacturers of insulin, a crucial medication for diabetes. Over decades of fierce competition, they had become leading providers of insulin and related pharmaceutical products. Novo Industri had been pursuing an internationally oriented strategy from the outset, and by 1936 was supplying insulin to 40 countries. A significant step in the internationalization of the company was a major push into the U.S. market in 1979. At the time, Food and Drug Administration (FDA) regulations required Novo Industri to replicate its clinical studies in the United States to obtain the approval of the marketing of their new products. In 1989, the two companies merged and, in 2000, the merged company spun off the enzyme business “Novozymes.”

In 2008, Novo Nordisk presents itself as a focused company within the healthcare industry and a world leader in diabetes care. It claims the broadest and most innovative diabetes product portfolio in the industry, including the most advanced insulin delivery systems. In addition, Novo Nordisk holds leading positions in areas such as haemostasis management, growth hormone therapy and hormone replacement therapy. Sales reached DKKr41.8 billion (about US\$8 billion) in 2007, of which DKKr30.5 billion were in diabetes care and DKKr11.4 billion were in biopharmaceuticals.

Innovation is considered pivotal to the success of Novo Nordisk, as it was to its predecessor companies. Continuous innovations allow the development of more refined, and thus more effective, insulin preparations, and new delivery systems, such as Novopen®, that facilitate the administration of the treatment, including self-administration by patients. In 2008, about 18 per cent of employees are working within research and development.

In 2008, Novo Nordisk holds market shares for insulin of about 56 per cent in Europe, 41 per cent in North America and 73 per cent in Japan and employs about 26,000 people, of whom 12,689 are located in Denmark, 3,411 in the rest of Europe, 3,940 in North America and the remainder in Asia Pacific and the rest of the world. Production facilities are located in six countries and products are marketed in 179 countries.

The shares of Novo Industri were first listed on the Copenhagen Stock Exchange in 1974 and on the London Stock Exchange in 1981 as the first Scandinavian company to be listed in London. In 2008, Novo Nordisk’s B shares are listed on the stock exchanges in both Copenhagen and London, while its American depositary receipts (ADRs) are listed on the New York Stock Exchange.

Novo Nordisk emphasizes corporate social responsibility as part of its image, pursuing a triple bottom line approach: environmental and social responsibility along with economic viability. This commitment is demonstrated through its values and its environmental and social responsibility policies that are reported on its website (see Appendix 1).

Critical milestones in Novo Nordisk’s ambition to be recognized as a leader of corporate sustainability include the publication in 1994 of its Environmental Report. It was the first company in Denmark and one of the first in the world to do so. This was followed in 1999 by the first annual Social Report. In 2001, Novo Nordisk established the World Diabetes Foundation, a charity aiming to improve diabetes care in developing countries, where diabetes is becoming an epidemic, just as it had in Europe and North America a few decades earlier.

² This section draws on the company website, www.novonordisk.com, and an undated (circa 2002) document, “Novo Nordisk History,” available via this website.

In recognition of its sustainability engagement, Novo Nordisk had been included in the Dow Jones Global Sustainability Indices, where it was ranked as “best in class” in the healthcare category in 2007. At home, Novo Nordisk is frequently ranked as having the most highly regarded corporate image by Danish magazines *Berlingske Nyhedsmagasinet*, *Børsen* and *Ingeniøren*.

New Medications: Development and Approval

Novo Nordisk, like other pharmaceutical and medical companies, heavily invests in the development of new medications offering more effective, safe and user-friendly treatments. New product development involves the creation of new drugs or modifications in their use, for instance, their dosage and the form of administration.

To bring new drugs or medical devices to market, they must be approved by the relevant authorities — the FDA in the United States and European Medicines Agency (EMA) in the European Union. The approval of drugs and medical devices requires proof of their efficacy and their safety. Efficacy refers to scientific evidence that the drug improves patients’ conditions as claimed by the manufacturer. Safety refers to the absence of substantive negative side-effects. Thus, to obtain approval, pharmaceutical companies have to provide scientific evidence that the drug improves the conditions of patients and is free of disproportional side effects.

This evidence has to be based on, among other data, clinical trials in which the drug has been tested on actual patients. The clinical trials are normally conducted in four stages. Phase 1 involves a small number of healthy volunteers and serves to assess the kinetic properties and tolerability of the drug. Phase 2 is performed on larger groups of patients to assess how well the drug works and to establish the doses that give the desired effect and to continue its safety assessment. Phase 3 trials often involve thousands of patients and aims to provide a definitive assessment of how effective and safe the drug is. All data generated in the three phases form an essential part of submissions to the regulatory authorities (FDA, EMA and their counterparts in other countries) for drug approval. With this approval, the drug can then be marketed for the approved indications. Further trials, in phase 4, may be required to obtain permission to extend the labelling of a drug to new indications (e.g. a different disease) or specific groups, such as children or pregnant women.

Phase 3 and 4 trials require a large number of patients with the specific disease that the drug is to improve. A typical approval process conducted by Novo Nordisk might require six to eight different phase 3 trials with different patient groups or combinations of the drug component, each involving about 400 to 800 patients. Such trials are often conducted as multinational studies involving up to 15 countries. With increasing requirements for patient exposure for approval and increasing numbers of drugs being tested, the recruitment of patients is often a major challenge. Typically, trials are conducted at multiple hospitals that all must follow the same trial protocol to ensure the consistency of data and compliance with existing “good clinical practice” (GCP) guidelines. Multi-site trials also facilitate the recruitment of patients with diverse backgrounds, for instance, different ethnicities and diets, while helping to demonstrate their universal properties. Doctors and nurses, but not patients, are normally paid for this work and hospitals often find it attractive to participate in trials that allow access to new medications and front line research. Clinical trials, especially phase 3, are a major cost factor in the development of new medications and they often take many years to conduct (on average eight years).

In the early 2000s, major pharmaceutical industries increasingly moved parts of their trials, especially phases 3 and 4, to countries outside their traditional areas of operations, especially to Eastern Europe, South America, India and China. Hospitals in these areas provide access to qualified medical staff and larger numbers of patients with the specific conditions, while potentially being able to administer a trial at lower costs. Moreover, the efficacy of drugs may also vary across contexts, for instance due to genetic, dietary, climatic or other environmental conditions. In such cases, multi-site trials help to establish the efficacy of medications across contexts. Some countries, such as Japan, India and China, in fact require that trials are at least in part conducted locally to approve a new medication in the respective countries. However, the conduct of clinical trials in these areas also raises a range of ethical issues.

Ethical Issues in Medical Research³

Ethical issues in the pharmaceutical industry have received considerable media attention over several decades, as the industry has failed to live up to the expectations of some interest groups. In particular, clinical trials raise a number of widely recognized issues. Medical professionals, and with them many NGOs and media, focus on the medical ethics grounded in the Hippocratic oath that commits doctors to treat each patient to the best of their abilities, never to cause intentional harm and to maintain patient confidentiality. Scientists and approval authorities have been concerned about the scientific rigor of the tests to provide solid evidence of the effects of a new drug, and thus to protect potential future users of the drug. At the same time, pharmaceutical companies have to operate with limited financial resources and to satisfy shareholders and thus cannot spend more resources than expected future revenues would justify. Accordingly, the industry has been accused of performing trials in developing countries with lower attention to ethical principles — “ethical bribing,” with patients acting as guinea pigs that do not understand and/or care about the risk involved but just want to get free medication, and with investigators not meeting the competence requirements, etc. Allegedly, all this just serves to generate documentation for compounds that are to be sold only in developed countries.

Medical (Hippocratic) ethics concern primarily the individual patients who are participating in any experiment. The relationship between the doctor and the subject participating in a trial is thus governed by the doctor’s responsibility to care for his or her patient. Past incidences where this principle had been violated continue to affect popular perceptions of medical research. Most infamously, the Tuskegee syphilis study left 400 impoverished and unwitting African-American men in Macon County, Alabama, untreated to study how they developed the disease — an experiment initiated in 1932 and terminated only in the 1970s.

To prevent such scandals, professional medical organizations have developed guidelines and principles of ethics to guide their research, notably the Helsinki Declaration of the World Medical Association (see Appendix 2). These widely accepted ethical principles aim to protect subjects, e.g., patients, participating in such research. These include:

- **Voluntary informed consent:** Each patient has to agree voluntarily to participate in the research based on being fully informed about the purposes of the study and potential risks for the individual. Sponsors and local site investigators thus normally write an “informed consent” document that informs potential subjects of the true risks and potential benefits, which is signed by each patient or their legal guardian before any trial procedure.

³ This section draws in particular on E.J. Emanuel, D. Wendler and C. Grady, “What makes clinical research ethical?” *Journal of the American Medical Association*, 283:20, 2000, pp. 2701-2711, and Michael A. Santoro and Thomas M. Gorrie, *Ethics and the Pharmaceutical Industry*, Cambridge University Press, 2005.

- Respect of patients: The privacy of the subject should be protected and they should be free to withdraw from the experiment at any time without reasoning. The doctor's professional responsibility to the patient should take precedence over any other considerations.
- Independent review: Any medical and pharmacological research has to be assessed on its scientific merits and ethicality by an independent review board (IRB) that is independent from those involved in or sponsoring the research.

Scientific ethics are concerned about the validity of the results of the scientific inquiry and thus the methodological rigour of the study. Thus, a study has to use valid measurements and statistical techniques and samples that are unbiased and sufficiently large that they can generate trustworthy and valid results.

Such scientific rigour is important to anyone who may, in the future, use an approved drug or medical device. Awareness of the need for rigorous tests prior to launching new medications had been triggered by various scandals of the 1960s, notably the Thalidomide scandal involving a pain killer used by women to ease sleep problems and pregnancy sickness. Due to side effects of this medication, thousands of children worldwide were born with incomplete arms or legs, before the drug was withdrawn. In consequence to this and other scandals, the licensing and approval procedures for drugs have been tightened to ensure that only drugs with scientifically proven efficacy and safety are marketed.

Ethical businesses have to balance activities done in the interest of the wider society with their pursuit of profits. The late Nobel prize-winning economist Milton Friedman famously declared that the primary social responsibility of business is to make profits.⁴ Under efficient markets, which he firmly believed in, this would generate the most mutually beneficial outcome. Thus, he argued, firms ought to give precedence to shareholders over any other interest groups.

Others argue that firms should engage in corporate social responsibility because it can be expected to benefit their bottom line in the long run, for instance through reputation effects. Yet others argue that firms have an intrinsic, normative responsibility to use their influence to do good for society and to aspire to the highest moral standards, independent of the profit motive. However, even so, their financial resources will be limited. Like organizations in the governmental or non-profit sector, businesses have to make critical decisions about how best to use their scarce resources.

Ethics of Placebo Experiments⁵

Particular concerns have arisen for placebo trials, that is, trials where a control group of patients receives a treatment without any active ingredient for the disease. The purpose of placebo trials is, normally, to provide evidence of product efficacy by showing statistically significant improvements of the conditions of patients receiving the active treatment, compared to those receiving a placebo treatment.

Placebo trials are especially important for diseases that are affected by the so-called placebo effect, that is, patients' conditions improving because of the positive effect of receiving a form of treatment rather than the specific medication. This has been shown to be quite substantive, for instance, for schizophrenia and

⁴ M. Friedman, "The social responsibility of business is to increase profits," *The New York Times Magazine*, September 13, 1970; reprinted in K.E. Meyer, *Multinational Enterprises and Host Economies*, Elgar, Cheltenham, 2009.

⁵ This section draws on contemporary discussions in the medical literature, in particular E.J. Emanuel and F.G. Miller, "The ethics of placebo-controlled trials e- A middle ground," *New England Journal of Medicine*, 345:12, 2001, pp. 915-919, and R. Temple and S.S. Ellenberg, "Placebo-controlled trials and active-controlled trials in the evaluation of new treatments," *Annals of Internal Medicine*, 133:6, 2000, pp. 455-463.

other psychiatric conditions. Both American and European authorities thus often require placebo trials as prerequisite for the approval of new medications.

Alternatives to placebo trials include the use of active controls, in which the control groups receive a previously marketed medication with known properties. Yet these types of trials are often not sufficient to provide the required rigorous evidence regarding the efficacy of the medication.⁶ Placebo trials may create risks for patients in the placebo group, in particular when patients are denied a treatment that is known to improve their condition. The Helsinki Declaration therefore requires avoiding placebo experiments unless very special reasons require them or no alternative treatment of the illness is available (see Appendix 2, item 29). Ethics review boards have become very restrictive in permitting placebo trials. There have been arguments from some groups that one reason for the pharmaceutical industry to place studies in developing countries is the possibility of performing placebo trials that otherwise can be difficult to get approval for in developed countries.

Novo Nordisk generally avoids placebo trials. Usually, they are used only in phase 1 trials in healthy volunteers when new drug candidates are being developed. These trials are normally located near its main research centres in Europe and rarely in non-Western countries.

Media Spotlights

In February 2008, a report from the Dutch NGO SOMO raised public awareness of placebo trials conducted by major pharmaceutical companies in developing countries.⁷ The report was critical of trials that had been submitted to the FDA and the EMEA for drug approval. Its primary concern was that key information about ethical aspects of these clinical trials was not available to it as an external observer and it found incidences where patients suffered serious harm after receiving a placebo in a trial.

The report focused on three case studies of clinical trials for recently approved drugs conducted in Eastern Europe and Asia, based on publicly available information. It concluded that:

Trial subjects in these countries are more vulnerable and their rights are less secured than in high income countries. Conditions such as poverty, illiteracy, poor health systems and inadequate research ethics committees result in international ethical standards not being met. Current EU legislation requires that results from unethical clinical trials . . . not be accepted for marketing authorization. With three case studies on recently approved drugs in the EU (Abilify, Olmetec, and Seroquel), SOMO demonstrates that this principle is being violated. European authorities devote little to no attention to the ethical aspects of the clinical trials submitted, and they accept unethical trials as well as trials of poor quality.⁸

⁶ An active-control trial infers efficacy from non-significant differences of performance compared to the active-control drug. Such non-significance, however, can be caused by a number of other influences. Moreover, this test is problematic if the active-control drug is subject to large placebo effects varying with study designs. On the merits and concerns of active-control trials, see e.g., Temple and Ellenberg, "Placebo-controlled trials and active-controlled trials in the evaluation of new treatments," 2000, and B.T. Walsh, S.N. Seidman, R. Sysko and M. Gould, "Placebo response in studies of major depression: Variable, substantial and growing," *Journal of the American Medical Association*, 287:14, 2002, pp. 1840-1847.

⁷ I. Schipper and F. Weyzing, "Ethics for Drug Testing in Low and Middle Income Countries: Considerations for European Market Authorisation," *Stichting Onderzoek Multinationale Ondernemingen (SOMO)*, 2008, http://somo.nl/publications-en/Publication_2472, accessed October 2008.

⁸ *Ibid*, abstract on the cover page.

In its conclusions, the report alleges that local regulation and the enforcement of ethical principles are less strict, partly because local independent review boards are less qualified and partly because they are less keen on restricting what is potentially a revenue earner. The authors thus advocate global harmonization of ethical criteria along the principles currently used by ethics committees in Europe: “. . . there must be no discrepancy between the ethical criteria used to approve research protocols in Western Europe and in low and middle income economies to avoid the creation of ‘easy countries.’”⁹

The media picked up, in particular, the case of a schizophrenic patient committing suicide while participating in a trial of the anti-schizophrenia medicine Seroquel by Astra-Zeneca. Moreover, media reported that 10 per cent of recipients in the placebo group had to be hospitalized because of worsening conditions. Careful reading of the original report suggests that 8.3 per cent (p. 64) of a group of 87 patients (p. 62) were affected, which adds up to seven persons. No assessment of the likelihood of such incidents under alternative medication available at the time had been included in the report.

Concerns have also been raised by the Danish National Committee on Biomedical Research Ethics.¹⁰ In particular, the committee criticizes the industry for not accepting the committee’s offer to provide independent ethical reviews before submitting to local ethics committees as a service to the industry. The chairperson for the committee, Johannes Gaub, chief medical officer at Odense Hospital, told the media:

Like production companies locate their factories in low wage areas, the medical industry is outsourcing its scientific experiments in the same way. The costs of conducting medical trials in developing countries are only a fraction of what they are in the West because of the low wages . . . In the USA it costs about DKr 150,000 to move one patient through a trial. In Denmark, it costs DKr 80,000. I don’t really know the price in developing countries, but it is a fraction of that.

Gaub also rejects the concern of the industry that hospitals in Denmark would not be able to conduct trials of the necessary scale, given the growing requirements worldwide to provide clinical trial data for approvals around the world:

We have considerable spare capacity in Denmark. Despite the high costs we have a well-functioning health system. We have data about patients because of our national identity number system, and there are many clinical researchers in the hospitals who would be happy to participate in the trials of new medications . . . It is actually worrying that we do not receive more applications in Denmark. We need clinical research to maintain the high level of health science that we so far have had in the country.¹¹

Danish politicians also joined the debate. In a statement to the health committee of the national parliament, the minister for health emphasized that E.U. regulation for the approval of new medicines requires that trials conducted outside the European Union have been implemented in accordance with the European Union’s own rules as well as with ethical principles such as the Helsinki Declaration. The minister thus concluded:

⁹ I. Schipper and F. Weyzing, “Ethics for Drug Testing in Low and Middle Income Countries: Considerations for European Market Authorisation,” 2008, p. 68.

¹⁰ For further information on the Danish National Committee on Biomedical Research Ethics, see www.cvk.im.dk/cvk/site.aspx?p=119.

¹¹ Both citations are from B. Erhardtsen, “Medicinalindustrien dropper frivillig etisk blåstempling,” *Berlingske Tidende*, April 5, 2008, Inland section, pp. 6-7 (case author’s translation).

I find no reason to take initiatives to constrain research projects by the Danish medical industry outside the EU. In this context, I consider it important to emphasize that all clinical trials that shall be used as a basis for applications for approval of marketing of a medication in the EU must comply with the EU's laws on good clinical practice and the ethical principles regarding medical research with human subjects.¹²

Other politicians also joined the debate. For example, Member of Parliament Birgitte Josefsen (V)¹³ urged Danish pharmaceutical companies to hold the ethical flag high: "The medical industry ought to be very careful about whom they use as test persons. That should be people who have resources to say 'no.' A poor Indian mother with three children is not the right one to become a test person."¹⁴

Novo Nordisk's Position on Clinical Trials

Anders Dejgaard is pondering the complexity of the ethical issues. As corporate sustainability features highly on Novo Nordisk's agenda, the ethically appropriate handling of clinical trials is important to the company. It conducts clinical trials globally to test the safety and efficacy of new drug candidates in order to obtain global marketing authorization. These trials always follow a common protocol and thus the same standards at all trials sites. Trials sites are selected based on a variety of criteria, including the quality of regulatory authorities, ethical review processes and medical practices. Moreover, drugs have to be tested on the types of patients who will later become users of the drug and trial subjects should have access to the drugs after the process has been completed. In addition, Novo Nordisk will only conduct trials in countries where it has affiliates with the necessary competence to arrange and monitor the trials. In 2008, these criteria were met in about 65 countries worldwide.

Novo Nordisk has adapted the global guidelines and recommendations by all the professional bodies and publishes its policies on clinical trials on its website (see Appendix 3). This includes enhanced global exposure of investigated products through its own website as well as websites sponsored by the FDA (see Appendix 3). Novo Nordisk conducts research in therapies that require global trials and the inclusion of different ethnic populations. The company also anticipates a need to increase the number of clinical trials due to an expanding pipeline and more extensive global and local regulatory requirements. Its ethical principles and standard operating procedures, which apply globally, are designed to ensure due respect for the safety, rights, integrity, dignity, confidentiality and well-being of all human beings participating in Novo Nordisk-sponsored trials. Novo Nordisk is auditing 10 per cent of all trials, while at the same time the American and European authorities, FDA and EMEA, are making random checks of about one per cent of Novo Nordisk's clinical trials. These random checks have never identified ethical problems in clinical trials in developing countries. Since trials are normally conducted in multiple countries, the same standards are applied everywhere, for both ethical and scientific reasons (consistency of results).

At the same time, Dejgaard is irritated about the request for an additional ethics approval by the Danish National Committee on Biomedical Research Ethics. He estimates that it would add three months to the preparation of each new trial. In his own experience, the ethical reviews in those locations he worked in are as rigorous as in Western countries and he does not recognize an added benefit, as the Danish committee

¹² J.K. Nielsen, *Besvarlse af spørgsmål nr. 20 (alm. del) som Folketingets Sundhedsudvalg har stillet til indenrigs- og sundhedsministeren* (Written reply to a question in the health committee of the Danish parliament), January 9, 2008. (Archives of the Danish government: Indenrigs og Sundhedsministeriet, Lægemiddelkontoret, J.nr. 2007-13009-599, Sagsbeh: nhj) (case author's translation).

¹³ (V) refers to Venstre, one of the parties of the minority government at the time.

¹⁴ Berlingske Tidende website, www.berlingske.dk/article/20080403/danmark/704030057, April 3, 2008, accessed October 2008 (case author's translation).

would be no better at assessing a trial than a local ethics committee. On the contrary, he finds the suggestion more appropriate for a colonial empire. Moreover, specific local issues, such as ethnic or religious minorities, would be better understood by local committees.

Yet various issues come to mind. Is Novo Nordisk doing its research and development in an appropriate manner or are there issues that could be done better in view of Novo Nordisk's triple bottom line commitments? Are Novo Nordisk's standard operating procedures being properly implemented in all developing countries that participate in the programs and how is such compliance to be monitored? How should Novo Nordisk manage its simultaneous relationships with various regulatory authorities, independent review boards at various sites and with the Danish National Committee on Biomedical Research Ethics?

Most pressing is the decision on how to handle the journalist. Should he meet her in person, send a public relations person, or not meet at all and reply in writing, citing the corporate website? If he is to meet her, what should be the key messages that he should get across and how should he prepare himself for any questions she might raise during the meeting?

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Appendix 1

CORPORATE SUSTAINABILITY AT NOVO NORDISK (EXTRACTS)

At Novo Nordisk, we refer to corporate sustainability as companies' ability to sustain and develop their business in the long-term perspective, in harmony with society. This implies a more inclusive view of business and its role; one in which engagements with stakeholders are not just used to legitimise corporate decisions, but rather the foundation for how it conducts and grows its business. It is about innovation, opportunity and planning for the long term.

The Triple Bottom is the principle behind our way of doing business. The company's Articles of Association state that it 'strives to conduct its activities in a financially, environmentally and socially responsible way.' This is a commitment to sustainable development and balanced growth, and it has been built into corporate governance structures, management tools and methods of assessing and rewarding individuals' performance...

The stakeholder dimension: Novo Nordisk needs to stay attuned to emerging trends and 'hot issues' on the global agenda in order to respond and to contribute to the debate. Stakeholder engagement is an integrated part of our business philosophy. We have long-standing engagements with stakeholders that are vital for building trust and understanding of a variety of issues. By involving stakeholders in the decision-making processes, decisions are better founded and solutions more likely to succeed. Stakeholders are defined as any individual or group that may affect or be affected by a company's activities.

Translating commitment to action: Corporate sustainability has made a meaningful difference to our business, and we believe it is a driver of our business success. This is best illustrated in three examples:

Business ethics: Surveys indicate that ethical behaviour in business is today the number one driver of reputation for pharmaceutical companies. Any company that is not perceived by the public as behaving in an ethical manner is likely to lose business, and it takes a long time to regain trust. While the Novo Nordisk Way of Management is a strong guide to our behaviour, we decided we needed more detailed guidance in the area of business ethics. In 2005 we therefore framed a new business ethics policy, in line with universally accepted high standards, backed by a set of procedures. Since then we have trained managers and employees, held workshops and offered e-learning on the new policy.

Climate change: We need to act to put a brake to human-induced climate change. While the implications of climate change pose major business risks, there are also opportunities. We have partnered with the WWF [World Wildlife Fund] in the Climate Savers programme and set an ambitious target to achieve a 10% reduction in our company's CO2 emissions by 2014, compared with 2004 emission levels. This will occur through optimised production, energy savings, and greater use of renewable energy supplies.

The diabetes pandemic: Today, diabetes is recognised as a pandemic. Novo Nordisk responds to this major societal challenge by working in partnerships with many others to rally the attention of policy-makers and influencers to change diabetes. We have made a promise of **Changing Diabetes®** and have framed a **strategy for inclusive access to diabetes care**. We established the **World Diabetes Foundation**, and have made several initiatives to advocate for change and build evidence of diabetes developments. **The National Changing Diabetes® programme** and **DAWN** are examples of education and awareness programmes implemented by Novo Nordisk affiliates in their respective countries. Our **Changing Diabetes® Bus** that promotes Novo Nordisk's global Changing Diabetes® activities had reached 86,000 people by the end of 2007 during its world tour. Its primary goal is to support the **UN Resolution on diabetes**, which was passed in December 2006.

Source: www.novonordisk.com, accessed November 2008.

Appendix 2**HELSINKI DECLARATION OF THE WORLD MEDICAL ASSOCIATION
(EXCERPTS)**

10. It is the duty of the physician in medical research to protect the life, health, privacy, and dignity of the human subject.

13. The design and performance of each experimental procedure involving human subjects should be clearly formulated in an experimental protocol. This protocol should be submitted for consideration, comment, guidance, and where appropriate, approval to a specially appointed ethical review committee, which must be independent of the investigator, the sponsor or any other kind of undue influence. This independent committee should be in conformity with the laws and regulations of the country in which the research experiment is performed. The committee has the right to monitor ongoing trials. The researcher has the obligation to provide monitoring information to the committee, especially any serious adverse events. The researcher should also submit to the committee, for review, information regarding funding, sponsors, institutional affiliations, other potential conflicts of interest and incentives for subjects.

14. The research protocol should always contain a statement of the ethical considerations involved and should indicate that there is compliance with the principles enunciated in this Declaration.

15. Medical research involving human subjects should be conducted only by scientifically qualified persons and under the supervision of a clinically competent medical person. The responsibility for the human subject must always rest with a medically qualified person and never rest on the subject of the research, even though the subject has given consent.

16. Every medical research project involving human subjects should be preceded by careful assessment of predictable risks and burdens in comparison with foreseeable benefits to the subject or to others. This does not preclude the participation of healthy volunteers in medical research. The design of all studies should be publicly available.

17. Physicians should abstain from engaging in research projects involving human subjects unless they are confident that the risks involved have been adequately assessed and can be satisfactorily managed. Physicians should cease any investigation if the risks are found to outweigh the potential benefits or if there is conclusive proof of positive and beneficial results.

18. Medical research involving human subjects should only be conducted if the importance of the objective outweighs the inherent risks and burdens to the subject. This is especially important when the human subjects are healthy volunteers.

19. Medical research is only justified if there is a reasonable likelihood that the populations in which the research is carried out stand to benefit from the results of the research.

20. The subjects must be volunteers and informed participants in the research project.

21. The right of research subjects to safeguard their integrity must always be respected. Every precaution should be taken to respect the privacy of the subject, the confidentiality of the patient's information and to minimize the impact of the study on the subject's physical and mental integrity and on the personality of the subject.

Appendix 2 (continued)

22. In any research on human beings, each potential subject must be adequately informed of the aims, methods, sources of funding, any possible conflicts of interest, institutional affiliations of the researcher, the anticipated benefits and potential risks of the study and the discomfort it may entail. The subject should be informed of the right to abstain from participation in the study or to withdraw consent to participate at any time without reprisal. After ensuring that the subject has understood the information, the physician should then obtain the subject's freely-given informed consent, preferably in writing. If the consent cannot be obtained in writing, the non-written consent must be formally documented and witnessed.

23. When obtaining informed consent for the research project the physician should be particularly cautious if the subject is in a dependent relationship with the physician or may consent under duress. In that case the informed consent should be obtained by a well-informed physician who is not engaged in the investigation and who is completely independent of this relationship.

29. The benefits, risks, burdens and effectiveness of a new method should be tested against those of the best current prophylactic, diagnostic, and therapeutic methods. This does not exclude the use of placebo, or no treatment, in studies where no proven prophylactic, diagnostic or therapeutic method exists.

Note of clarification on paragraph 29 of the WMA Declaration of Helsinki

The WMA hereby reaffirms its position that extreme care must be taken in making use of a placebo-controlled trial and that in general this methodology should only be used in the absence of existing proven therapy. However, a placebo-controlled trial may be ethically acceptable, even if proven therapy is available, under the following circumstances:

- Where for compelling and scientifically sound methodological reasons its use is necessary to determine the efficacy or safety of a prophylactic, diagnostic or therapeutic method; or
- Where a prophylactic, diagnostic or therapeutic method is being investigated for a minor condition and the patients who receive placebo will not be subject to any additional risk of serious or irreversible harm.

All other provisions of the Declaration of Helsinki must be adhered to, especially the need for appropriate ethical and scientific review.

30. At the conclusion of the study, every patient entered into the study should be assured of access to the best proven prophylactic, diagnostic and therapeutic methods identified by the study.

Note of clarification on paragraph 30 of the WMA Declaration of Helsinki

The WMA hereby reaffirms its position that it is necessary during the study planning process to identify post-trial access by study participants to prophylactic, diagnostic and therapeutic procedures identified as beneficial in the study or access to other appropriate care. Post-trial access arrangements or other care must be described in the study protocol so the ethical review committee may consider such arrangements during its review.

Source: www.wma.net/e/policy/b3.htm, accessed October 2008.

Appendix 3

CLINICAL TRIALS: NOVO NORDISK'S POSITION

- Clinical trials sponsored by Novo Nordisk will always be conducted according to the Helsinki Declaration, which describes human rights for patients participating in clinical trials, and similar international ethical guidelines such as the Nuremberg code, the Belmont report and CIOMMS, and the International Conference of Harmonisation (ICH) guidelines for current good clinical practice (cGCP).
- The above guidelines and regulations are the foundation for our clinical Standard Operating Procedures (SOPs) including the SOP on the 'principles of clinical trials'. These standards are laid out to ensure the safety, rights, integrity, confidentiality and well-being of persons involved in Novo Nordisk trials globally.
- Novo Nordisk will apply the same procedures wherever we sponsor clinical trials. This means that all subjects enrolled in Novo Nordisk trials are protected by the same rights, high ethical standards and regulations irrespective of location of the study.
- The interest and well-being of the trial subject should always prevail over the interest of science, society and commerce.
- Novo Nordisk will not conduct clinical trials for drug development in countries where we do not intend to market the investigational drug. In any country where we do undertake clinical trials we will ensure that a proper internal organisation and a proper regulated external environment exist.
- Clinical trials should only be done if they can be scientifically and medically justified, and all Novo Nordisk-sponsored trials should be based on sound scientific methodology described in a clear and detailed protocol. Placebo will only be used as comparator when scientifically and ethically justified.
- No trial activity in Novo Nordisk-sponsored trials will start before approval is obtained from external local ethics committees and health authorities.
- We will always ensure that investigators involved in Novo Nordisk clinical trials are skilled in the therapeutic area and are trained in GCP. No procedure involving a person undergoing clinical trial activities will take place before the appropriate freely given informed consent is obtained based on proper information on potential risk of participation in the trial. A patient can at any time withdraw from a clinical trial without giving any reason. In cases where trial subjects are incompetent, physically or mentally incapable of giving consent, or if the person is a minor, Novo Nordisk will follow local regulations for obtaining consent.
- Products used in Novo Nordisk-sponsored clinical trials will be manufactured and controlled according to international and local regulations and laws. Novo Nordisk will conduct frequent site monitoring to ensure that the study is executed according to the study protocol, and that data used in statistical analysis and reporting reflects the data obtained from the involved patients during the trial. Safety information from any Novo Nordisk trial will be monitored on a continuous basis and appropriate actions will be taken if risks of the investigational product outweigh the potential benefits.
- Patients participating in Novo Nordisk-sponsored clinical trials will always be offered best available and proven treatment after study termination. The treatment will be offered at the discretion of the responsible physician. If study medication is not marketed the responsible physician can apply for medication on a named patient basis. Post-study medication will be described in the protocol and informed consent.
- Novo Nordisk will ensure proper indemnification of trial subjects in case a trial product or procedures in a Novo Nordisk-sponsored trial cause bodily harm to a trial subject.
- Novo Nordisk strives to have all clinical trial results published according to accepted international guidelines, and we will always ensure transparency of our studies by publishing protocol synopses on the external website: www.clinicaltrials.gov. Study results from trials involving marketed drugs can be accessed via www.clinicalstudyresults.org. Furthermore, Novo Nordisk has its own online repository for clinical trials activities: novonordisk-trials.com. Novo Nordisk is collating all information about bioethics in the R&D area on www.novonordisk.com/R&D/bioethics.

Source: www.novonordisk.com, accessed November 2008.